

Yuanqing Ma

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EDUCATION

The Chinese University of Hong Kong, Shenzhen

Bachelor of Science in Computer Science and Engineering

Shenzhen, China

Sept. 2024 – Present

- **CGPA:** 3.940/4.000 (Major Ranking: **4/233**)
- **Relevant Coursework:** Data Structures, Matrix Computations, Machine Learning, Reinforcement Learning.
- **Research Interests:** Reinforcement Learning, Agentic RL, In-Context RL

RESEARCH EXPERIENCE

Research Assistant

Sept. 2025 – Present

The Chinese University of Hong Kong, Shenzhen

Advised by Prof. Chenjun Xiao

- Reproduced and extended **Algorithm Distillation (AD)**: Distilled RL algorithms into Transformer architectures for zero-shot generalization. [\[Code on GitHub\]](#)
- Designed a **context compression mechanism** for AD, reducing inference context length to 20–25% while preserving performance and significantly improving efficiency.
- Built a high-performance multiprocessing **PPO data collection pipeline** in PyTorch to gather trajectories and implemented custom Transformer policies for Grid-World and Meta-World tasks.
- **Publication:** First-author paper on In-Context RL currently under review at **ICML 2026**.

PROJECTS

Self-Summarization Deep Research Agent | [\[GitHub\]](#)

Apr. 2026

- Built a self-summarizing research agent for BrowseComp-Plus that performs iterative search, get document, and finish tool calls while automatically compressing context to avoid overflowing the context window.
- Implemented an RL training pipeline including offline vLLM rollout collection, reward-aligned trajectory extraction, and customized GRPO trainer.
- Developed configurable experiment launchers for benchmark runs and result analysis.

Autonomous AI Data Scientist Agent | [\[GitHub\]](#)

Feb. 2026

- Designed a multi-stage agentic framework that autonomously completes the ML lifecycle using local LLMs.
- Implemented an execution-feedback decision loop where the agent iteratively drafts code, analyzes traceback signals, and refines strategies through self-correction in a sandboxed environment.

Deep Reinforcement Learning Implementations (Course Projects) | [\[GitHub\]](#)

Feb. 2026

- **Imitation Learning:** Implemented Flow Matching vs. MSE for Action Chunking in Push-T environments.
- **Online RL:** Implemented major RL algorithms (e.g. GAE, DQN, SAC) on Atari and Mujoco tasks.
- **LLM RL:** Fine-tuned LLMs using GRPO and GR-REINFORCE for complex mathematical reasoning tasks.
- **Offline RL:** Developed and compared behavior-regularized, value-based, and flow-based offline RL methods (SAC+BC, IQL, FQL) on robotic control environments.
- Authored a literature review on In-context RL. [\[PDF\]](#)

TEACHING

Undergraduate Student Teaching Fellow (USTF)

Sept. 2025 – Present

- **MAT2041A (Linear Algebra):** Leading tutorials and grading for Spring 2026.
- **CSC1003 (Intro to Java):** Supported tutorial and practice sessions for Fall 2025.

HONORS & SCHOLARSHIPS

Academic Performance Scholarship (Class A) & Dean's List: Top-tier academic recognition (AY 2024–25).
Full-tuition Scholarship & Sin Wai Kin Special Admissions Scholarship (2024–Present).

TECHNICAL SKILLS

Languages: Python, Java, C, C++, RISC-V, LaTeX

Libraries: PyTorch, HuggingFace, NumPy, Stable-Baselines3, vLLM, SGLang, OpenRLHF

Tools & Environments: Linux (Ubuntu), Git, GitHub, Gymnasium, Meta-World

LEADERSHIP & ACTIVITIES

Shaw College Men's Soccer Team

Sept. 2024 – Present

Assistant Coach & Player

Shenzhen, China

- Gold Medal in the 2025 CUHK(SZ) Olympics and the Shaw Outstanding Student Organization Award.